

FixFlow ERP's strict four-step financial chain, state-machine gates, and generic attachment structure require specialized reporting to maintain cash accountability, operational throughput, and database integrity.

Financial & Cash Chain-of-Custody Audits

- **Spare Change Leakage & Float Exposure Audit:** Tracks disbursed transmittals where `spare_change_required > spare_change_returned` or where requisitions are stuck in FULFILLED awaiting Form 14 sign-off. Identifies outstanding cash held by purchasers or Front Desk revolving floats (`fdCodDisbursement`).
- **Canvass vs. Actual Purchase Variance Report:** Audits price drift by comparing Budget Officer estimates (Form 8 `item_price_catalog`) against Purchaser execution actuals (Form 13). Flags vendor categories consistently exceeding initial budget allocation.
- **Gate Violation & Exception Log:** Captures attempted state jumps blocked by database triggers (Migrations 0012–0015), such as attempts to mark transmittals RECEIVED prior to SENT, or attempts by Accounting to close requisitions without requester delivery sign-off (`requester_verification <> 'VERIFIED'`).

Operational & Workflow Analytics

- **Requisition Stage Aging & Bottleneck Report:** Measures mean lead times across stage gates (`PENDING_MANAGER` $\rightarrow$ `IN_CANVASSING` $\rightarrow$ `TRANSMITTAL_IN_PROGRESS` $\rightarrow$ `PURCHASING` $\rightarrow$ `DELIVERED`). Isolates operational delays caused by Manager approvals versus Accounting cash disbursement.
- **Partial Availability & Shortfall Analytics:** Tracks line items triggering `availability_hold` (Gate A) and logs requester resolution choices (`PROCEED_PARTIAL`, `CANCEL_REMAINING`, `WAIT_FULL`) to pinpoint vendor reliability issues and unfulfilled supply rates.
- **Asset Recurrence & Escalation Log:** Filters assets and locations with high rates of `CRITICAL_REOPEN_ESCALATED` tickets or frequent PMS service interventions (Forms 15–16) to inform replacement decisions and track technician performance.

System & Data Integrity Logs

- **Orphan Attachment & Storage Audit:** Scans the `attachments` table for `entity_id` values missing corresponding primary records in `job_orders` or `material_requisitions` (necessary because `attachments` lacks foreign key constraints), and cross-references active files in Supabase storage buckets.
- **Reference Sequence Continuity Audit:** Validates contiguous incrementation of `next_reference_number()` output across JO, MRS, TR, and TR-BATCH prefixes to detect failed transactions or uncommitted sequence increments.
- **Security & Privilege Escalation Audit:** Monitors all `user_role` changes, account status toggles (ACTIVE vs INACTIVE in Form 18), and session auth failures caught by `getServerUser()`.